

1.6 Exercises

Exercise 1.1: State space form for chemical reaction model

Consider the following chemical reaction kinetics for a two-step series reaction



We wish to follow the reaction in a constant volume, well-mixed, batch reactor. As taught in the undergraduate chemical engineering curriculum, we proceed by writing material balances for the three species giving

$$\frac{dc_A}{dt} = -r_1 \quad \frac{dc_B}{dt} = r_1 - r_2 \quad \frac{dc_C}{dt} = r_2$$

in which c_j is the concentration of species j , and r_1 and r_2 are the rates (mol/(time·vol)) at which the two reactions occur. We then assume some rate law for the reaction kinetics, such as

$$r_1 = k_1 c_A \quad r_2 = k_2 c_B$$

We substitute the rate laws into the material balances and specify the starting concentrations to produce three differential equations for the three species concentrations.

- Write the linear state space model for the deterministic series chemical reaction model. Assume we can measure the component A concentration. What are x , y , A , B , C , and D for this model?
- Simulate this model with initial conditions and parameters given by

$$c_{A0} = 1 \quad c_{B0} = c_{C0} = 0 \quad k_1 = 2 \quad k_2 = 1$$

Exercise 1.2: Distributed systems and time delay

We assume familiarity with the transfer function of a time delay from an undergraduate systems course

$$\overline{y}(s) = e^{-\theta s} \overline{u}(s)$$

Let's see the connection between the delay and the distributed systems, which give rise to it. A simple physical example of a time delay is the delay caused by transport in a flowing system. Consider plug flow in a tube depicted in Figure 1.12.

- Write down the equation of change for moles of component j for an arbitrary volume element and show that

$$\frac{\partial c_j}{\partial t} = -\nabla \cdot (c_j v_j) + R_j$$

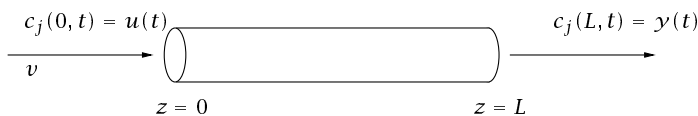


Figure 1.12: Plug-flow reactor.

in which c_j is the molar concentration of component j , v_j is the velocity of component j , and R_j is the production rate of component j due to chemical reaction.⁹

Plug flow means the fluid velocity of all components is purely in the z direction, and is independent of r and θ and, we assume here, z

$$v_j = v \delta_z$$

- (b) Assuming plug flow and neglecting chemical reaction in the tube, show that the equation of change reduces to

$$\frac{\partial c_j}{\partial t} = -v \frac{\partial c_j}{\partial z} \quad (1.46)$$

This equation is known as a hyperbolic, first-order partial differential equation. Assume the boundary and initial conditions are

$$c_j(z, t) = u(t) \quad 0 = z \quad t \geq 0 \quad (1.47)$$

$$c_j(z, t) = c_{j0}(z) \quad 0 \leq z \leq L \quad t = 0 \quad (1.48)$$

In other words, we are using the feed concentration as the manipulated variable, $u(t)$, and the tube starts out with some initial concentration profile of component j , $c_{j0}(z)$.

- (c) Show that the solution to (1.46) with these boundary conditions is

$$c_j(z, t) = \begin{cases} u(t - z/v) & vt > z \\ c_{j0}(z - vt) & vt < z \end{cases} \quad (1.49)$$

- (d) If the reactor starts out empty of component j , show that the transfer function between the outlet concentration, $y = c_j(L, t)$, and the inlet concentration, $c_j(0, t) = u(t)$, is a time delay. What is the value of θ ?

Exercise 1.3: Pendulum in state space

Consider the pendulum suspended at the end of a rigid link depicted in Figure 1.13. Let r and θ denote the polar coordinates of the center of the pendulum, and let $p = r\delta_r$ be the position vector of the pendulum, in which δ_r and δ_θ are the unit vectors in polar coordinates. We wish to determine a state space description of the system. We are able to apply a torque T to the pendulum as our manipulated variable. The pendulum has mass m , the only other external force acting on the pendulum is gravity, and we neglect friction. The link provides force $-t\delta_r$ necessary to maintain the pendulum at distance $r = R$ from the axis of rotation, and we measure this force t .

- (a) Provide expressions for the four partial derivatives for changes in the unit vectors with r and θ

$$\frac{\partial \delta_r}{\partial r} \quad \frac{\partial \delta_r}{\partial \theta} \quad \frac{\partial \delta_\theta}{\partial r} \quad \frac{\partial \delta_\theta}{\partial \theta}$$

- (b) Use the chain rule to find the velocity of the pendulum in terms of the time derivatives of r and θ . Do not simplify yet by assuming r is constant. We want the general result.

⁹You will need the Gauss divergence theorem and 3D Leibniz formula to go from a mass balance on a volume element to the equation of continuity.

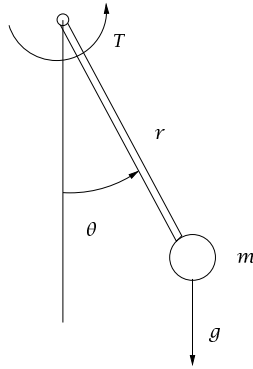


Figure 1.13: Pendulum with applied torque.

- (c) Differentiate again to show that the acceleration of the pendulum is

$$\ddot{\mathbf{p}} = (\ddot{r} - r\dot{\theta}^2)\delta_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\delta_\theta$$

- (d) Use a momentum balance on the pendulum mass (you may assume it is a point mass) to determine both the force exerted by the link

$$t = mR\dot{\theta}^2 + mg \cos \theta$$

and an equation for the acceleration of the pendulum due to gravity and the applied torque

$$mR\ddot{\theta} - T/R + mg \sin \theta = 0$$

- (e) Define a state vector and give a state space description of your system. What is the physical significance of your state. Assume you measure the force exerted by the link.

One answer is

$$\begin{aligned} \frac{dx_1}{dt} &= x_2 \\ \frac{dx_2}{dt} &= -(g/R) \sin x_1 + u \\ y &= mRx_2^2 + mg \cos x_1 \end{aligned}$$

in which $u = T/(mR^2)$

Exercise 1.4: Time to Laplace domain

Take the Laplace transform of the following set of differential equations and find the transfer function, $G(s)$, connecting $\bar{u}(s)$ and $\bar{y}(s)$, $\bar{y} = G\bar{u}$

$$\begin{aligned} \frac{dx}{dt} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \tag{1.50}$$

For $x \in \mathbb{R}^n$, $y \in \mathbb{R}^p$, and $u \in \mathbb{R}^m$, what is the dimension of the G matrix? What happens to the initial condition, $x(0) = x_0$?

Exercise 1.5: Converting between continuous and discrete time models

Given a prescribed $u(t)$, derive and check the solution to (1.50). Given a prescribed $u(k)$ sequence, what is the solution to the discrete time model

$$\begin{aligned}x(k+1) &= \tilde{A}x(k) + \tilde{B}u(k) \\y(k) &= \tilde{C}x(k) + \tilde{D}u(k)\end{aligned}$$

- Compute $\tilde{A}, \tilde{B}, \tilde{C}$, and $\tilde{D}$ so that the two solutions agree at the sample times for a zero-order hold input, i.e., $y(k) = y(t_k)$ for $u(t) = u(k)$, $t \in (t_k, t_{k+1})$ in which $t_k = k\Delta$ for sample time Δ .
- Is your result valid for A singular? If not, how can you find $\tilde{A}, \tilde{B}, \tilde{C}$, and $\tilde{D}$ for this case?

Exercise 1.6: Continuous to discrete time conversion for nonlinear models

Consider the autonomous nonlinear differential equation model

$$\begin{aligned}\frac{dx}{dt} &= f(x, u) \\x(0) &= x_0\end{aligned}\tag{1.51}$$

Given a zero-order hold on the input, let $s(t, u, x_0)$, $0 \leq t \leq \Delta$, be the solution to (1.51) given initial condition x_0 at time $t = 0$, and constant input u is applied for t in the interval $0 \leq t \leq \Delta$. Consider also the nonlinear discrete time model

$$x(k+1) = F(x(k), u(k))$$

- What is the relationship between F and s so that the solution of the discrete time model agrees at the sample times with the continuous time model with a zero-order hold?
- Assume f is linear and apply this result to check the result of Exercise 1.5.

Exercise 1.7: Commuting functions of a matrix

Although matrix multiplication does not commute in general

$$AB \neq BA$$

multiplication of functions of the same matrix do commute. You may have used the following fact in Exercise 1.5

$$A^{-1} \exp(At) = \exp(At) A^{-1}\tag{1.52}$$

- Prove that (1.52) is true assuming A has distinct eigenvalues and can therefore be represented as

$$A = Q\Lambda Q^{-1} \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

in which Λ is a diagonal matrix containing the eigenvalues of A , and Q is the matrix of eigenvectors such that

$$Aq_i = \lambda_i q_i, \quad i = 1, \dots, n$$

in which q_i is the i th column of matrix Q .

- (b) Prove the more general relationship

$$f(A)g(A) = g(A)f(A) \quad (1.53)$$

in which f and g are any functions definable by Taylor series.

- (c) Prove that (1.53) is true without assuming the eigenvalues are distinct.

Hint: use the Taylor series defining the functions and apply the Cayley-Hamilton theorem (Horn and Johnson, 1985, pp. 86-87).

Exercise 1.8: Finite difference formula and approximating the exponential

Instead of computing the exact conversion of a continuous time to a discrete time system as in Exercise 1.5, assume instead one simply approximates the time derivative with a first-order finite difference formula

$$\frac{dx}{dt} \approx \frac{x(t_{k+1}) - x(t_k)}{\Delta}$$

with step size equal to the sample time, Δ . For this approximation of the continuous time system, compute $\tilde{A}$ and $\tilde{B}$ so that the discrete time system agrees with the approximate continuous time system at the sample times. Comparing these answers to the exact solution, what approximation of $e^{A\Delta}$ results from the finite difference approximation? When is this a good approximation of $e^{A\Delta}$?

Exercise 1.9: Mapping eigenvalues of continuous time systems to discrete time systems

Consider the continuous time differential equation and discrete time difference equation

$$\begin{aligned} \frac{dx}{dt} &= Ax \\ x^+ &= \tilde{A}x \end{aligned}$$

and the transformation

$$\tilde{A} = e^{A\Delta}$$

Consider the scalar A case.

- What A represents an integrator in continuous time? What is the corresponding $\tilde{A}$ value for the integrator in discrete time?
- What A give purely oscillatory solutions? What are the corresponding $\tilde{A}$?
- For what A is the solution of the ODE stable? Unstable? What are the corresponding $\tilde{A}$?
- Sketch and label these A and $\tilde{A}$ regions in two complex-plane diagrams.

Exercise 1.10: State space realization

Define a state vector and realize the following models as state space models **by hand**. One should do a few by hand to understand what the Octave or MATLAB calls are doing. Answer the following questions. What is the connection between the poles of G and the state space description? For what kinds of $G(s)$ does one obtain a nonzero D matrix? What is the order and gain of these systems? Is there a connection between order and the numbers of inputs and outputs?

$$(a) \quad G(s) = \frac{1}{2s+1}$$

$$(b) \quad G(s) = \frac{1}{(2s+1)(3s+1)}$$

$$(c) \quad G(s) = \frac{2s+1}{3s+1}$$

$$(d) \quad y(k+1) = y(k) + 2u(k)$$

$$(e) \quad y(k+1) = a_1 y(k) + a_2 y(k-1) + b_1 u(k) + b_2 u(k-1)$$

Exercise 1.11: Minimal realization

Find minimal realizations of the state space models you found by hand in Exercise 1.10. Use Octave or MATLAB for computing minimal realizations. Were any of your hand realizations nonminimal?

Exercise 1.12: Partitioned matrix inversion lemma

Let matrix Z be partitioned into

$$Z = \begin{bmatrix} B & C \\ D & E \end{bmatrix}$$

and assume Z^{-1} , B^{-1} and E^{-1} exist.

(a) Perform row elimination and show that

$$Z^{-1} = \begin{bmatrix} B^{-1} + B^{-1}C(E - DB^{-1}C)^{-1}DB^{-1} & -B^{-1}C(E - DB^{-1}C)^{-1} \\ -(E - DB^{-1}C)^{-1}DB^{-1} & (E - DB^{-1}C)^{-1} \end{bmatrix}$$

Note that this result is still valid if E is singular.

(b) Perform column elimination and show that

$$Z^{-1} = \begin{bmatrix} (B - CE^{-1}D)^{-1} & -(B - CE^{-1}D)^{-1}CE^{-1} \\ -E^{-1}D(B - CE^{-1}D)^{-1} & E^{-1} + E^{-1}D(B - CE^{-1}D)^{-1}CE^{-1} \end{bmatrix}$$

Note that this result is still valid if B is singular.

(c) A host of other useful control-related inversion formulas follow from these results. Equate the (1,1) or (2,2) entries of Z^{-1} and derive the identity

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(DA^{-1}B + C^{-1})^{-1}DA^{-1} \quad (1.54)$$

A useful special case of this result is

$$(I + X^{-1})^{-1} = I - (I + X)^{-1}$$

(d) Equate the (1,2) or (2,1) entries of Z^{-1} and derive the identity

$$(A + BCD)^{-1}BC = A^{-1}B(DA^{-1}B + C^{-1})^{-1} \quad (1.55)$$

Equations (1.54) and (1.55) prove especially useful in rearranging formulas in least squares estimation.

Exercise 1.13: Perturbation to an asymptotically stable linear system

Given the system

$$x^+ = Ax + Bu$$

If A is an asymptotically stable matrix, prove that if $u(k) \rightarrow 0$, then $x(k) \rightarrow 0$.

Exercise 1.14: Exponential stability of a perturbed linear system

Given the system

$$x^+ = Ax + Bu$$

If A is an asymptotically stable matrix, prove that if $u(k)$ decreases exponentially to zero, then $x(k)$ decreases exponentially to zero.

Exercise 1.15: Are we going forward or backward today?

In the chapter we derived the solution to

$$\min_{w, x, y} f(w, x) + g(x, y) + h(y, z)$$

in which z is a fixed parameter using forward dynamic programming (DP)

$$\bar{y}^0(z)$$

$$\tilde{x}^0(z) = \bar{x}^0(\bar{y}^0(z))$$

$$\tilde{w}^0(z) = \bar{w}^0(\tilde{x}^0(\bar{y}^0(z)))$$

- (a) Solve for optimal w as a function of z using backward DP.
- (b) Is forward or backward DP more efficient if you want optimal w as a function of z ?

Exercise 1.16: Method of Lagrange multipliers

Consider the objective function $V(x) = (1/2)x'Hx + h'x$ and optimization problem

$$\min_x V(x) \tag{1.56}$$

subject to

$$Dx = d$$

in which $H > 0$, $x \in \mathbb{R}^n$, $d \in \mathbb{R}^m$, $m < n$, i.e., fewer constraints than decisions. Rather than partially solving for x using the constraint and eliminating it, we make use of the method of Lagrange multipliers for treating the equality constraints (Fletcher, 1987; Nocedal and Wright, 2006).

In the method of Lagrange multipliers, we augment the objective function with the constraints to form the Lagrangian function, L

$$L(x, \lambda) = (1/2)x'Hx + h'x - \lambda'(Dx - d)$$

in which $\lambda \in \mathbb{R}^m$ is the vector of Lagrange multipliers. The necessary and sufficient conditions for a global minimizer are that the partial derivatives of L with respect to x and λ vanish (Nocedal and Wright, 2006, p. 451), (Fletcher, 1987, p.198,236)

- (a) Show that the necessary and sufficient conditions are equivalent to the matrix equation

$$\begin{bmatrix} H & -D' \\ -D & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = - \begin{bmatrix} h \\ d \end{bmatrix} \quad (1.57)$$

The solution to (1.57) then provides the solution to the original problem (1.56).

- (b) We note one other important feature of the Lagrange multipliers, their relationship to the optimal cost of the purely quadratic case. For $h = 0$, the cost is given by

$$V^0 = (1/2)(x^0)' H x^0$$

Show that this can also be expressed in terms of λ^0 by the following

$$V^0 = (1/2)d' \lambda^0$$

Exercise 1.17: Minimizing a constrained, quadratic function

Consider optimizing the positive definite quadratic function subject to a linear constraint

$$\min_x (1/2)x' H x \quad \text{s.t. } Ax = b$$

Using the method of Lagrange multipliers presented in Exercise 1.16, show that the optimal solution, multiplier, and cost are given by

$$x^0 = H^{-1} A' (A H^{-1} A')^{-1} b$$

$$\lambda^0 = (A H^{-1} A')^{-1} b$$

$$V^0 = (1/2)b' (A H^{-1} A')^{-1} b$$

Exercise 1.18: Minimizing a partitioned quadratic function

Consider the partitioned constrained minimization

$$\min_{x_1, x_2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' \begin{bmatrix} H_1 & \\ & H_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

subject to

$$\begin{bmatrix} D & I \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = d$$

The solution to this optimization is required in two different forms, depending on whether one is solving an estimation or regulation problem. Show that the solution can be expressed in the following two forms if both H_1 and H_2 are full rank.

- Regulator form

$$V^0(d) = d' (H_2 - H_2 D (D' H_2 D + H_1)^{-1} D' H_2) d$$

$$x_1^0(d) = \tilde{K} d \quad \tilde{K} = (D' H_2 D + H_1)^{-1} D' H_2$$

$$x_2^0(d) = (I - D \tilde{K}) d$$

- Estimator form

$$V^0(d) = d' (D H_1^{-1} D' + H_2^{-1})^{-1} d$$

$$x_1^0(d) = \tilde{L} d \quad \tilde{L} = H_1^{-1} D' (D H_1^{-1} D' + H_2^{-1})^{-1}$$

$$x_2^0(d) = (I - D \tilde{L}) d$$

Exercise 1.19: Stabilizability and controllability canonical forms

Consider the partitioned system

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^+ = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u$$

with (A_{11}, B_1) controllable. This form is known as controllability canonical form.

- Show that the system is *not* controllable by checking the rank of the controllability matrix.
- Show that the modes x_1 can be controlled from any $x_1(0)$ to any $x_1(n)$ with a sequence of inputs $u(0), \dots, u(n-1)$, but the modes x_2 *cannot* be controlled from any $x_2(0)$ to any $x_2(n)$. The states x_2 are termed the uncontrollable modes.
- If A_{22} is stable the system is termed *stabilizable*. Although not all modes can be controlled, the uncontrollable modes are stable and decay to steady state.

The following lemma gives an equivalent condition for stabilizability.

Lemma 1.12 (Hautus lemma for stabilizability). *A system is stabilizable if and only if*

$$\text{rank} \begin{bmatrix} \lambda I - A & B \end{bmatrix} = n \quad \text{for all } |\lambda| \geq 1$$

Prove this lemma using Lemma 1.2 as the condition for controllability.

Exercise 1.20: Regulator stability, stabilizable systems, and semidefinite state penalty

- Show that the infinite horizon LQR is stabilizing for (A, B) stabilizable with $R, Q > 0$.
- Show that the infinite horizon LQR is stabilizing for (A, B) stabilizable and $R > 0, Q \geq 0$, and (A, Q) detectable. Discuss what happens to the controller's stabilizing property if Q is not positive semidefinite or (A, Q) is not detectable.

Exercise 1.21: Time-varying linear quadratic problem

Consider the time-varying version of the LQ problem solved in the chapter. The system model is

$$x(k+1) = A(k)x(k) + B(k)u(k)$$

The objective function also contains time-varying penalties

$$\min_{\mathbf{u}} V(x(0), \mathbf{u}) = \frac{1}{2} \left(\sum_{k=0}^{N-1} (x(k)' Q(k) x(k) + u(k)' R(k) u(k)) + x(N)' Q(N) x(N) \right)$$

subject to the model. Notice the penalty on the final state is now simply $Q(N)$ instead of P_f .

Apply the DP argument to this problem and determine the optimal input sequence and cost. Can this problem also be solved in closed form like the time-invariant case?

Exercise 1.22: Steady-state Riccati equation

Generate a random A and B for a system model for whatever $n (\geq 3)$ and $m (\geq 3)$ you wish. Choose a positive semidefinite Q and positive definite R of the appropriate sizes.

- (a) Iterate the DARE by hand with Octave or MATLAB until Π stops changing. Save this result. Now call the MATLAB or Octave function to solve the steady-state DARE. Do the solutions agree? Where in the complex plane are the eigenvalues of $A + BK$? Increase the size of Q relative to R . Where do the eigenvalues move?
- (b) Repeat for a singular A matrix. What happens to the two solution techniques?
- (c) Repeat for an unstable A matrix.

Exercise 1.23: Positive definite Riccati iteration

If $\Pi(k), Q, R > 0$ in (1.10), show that $\Pi(k-1) > 0$.

Hint: apply (1.54) to the term $(B' \Pi(k) B + R)^{-1}$.

Exercise 1.24: Existence and uniqueness of the solution to constrained least squares

Consider the least squares problem subject to linear constraint

$$\min_x (1/2) x' Q x \quad \text{subject to} \quad Ax = b$$

in which $x \in \mathbb{R}^n$, $b \in \mathbb{R}^p$, $Q \in \mathbb{R}^{n \times n}$, $Q \geq 0$, $A \in \mathbb{R}^{p \times n}$. Show that this problem has a solution for every b and the solution is unique if and only if

$$\text{rank}(A) = p \quad \text{rank} \begin{bmatrix} Q \\ A \end{bmatrix} = n$$

Exercise 1.25: Rate-of-change penalty

Consider the generalized LQR problem with the cross term between $x(k)$ and $u(k)$

$$V(x(0), \mathbf{u}) = \frac{1}{2} \sum_{k=0}^{N-1} (x(k)' Q x(k) + u(k)' R u(k) + 2x(k)' M u(k)) + (1/2) x(N)' P_f x(N)$$

- (a) Solve this problem with backward DP and write out the Riccati iteration and feedback gain.
- (b) Control engineers often wish to tune a regulator by penalizing the rate of change of the input rather than the absolute size of the input. Consider the additional positive definite penalty matrix S and the modified objective function

$$V(x(0), \mathbf{u}) = \frac{1}{2} \sum_{k=0}^{N-1} (x(k)' Q x(k) + u(k)' R u(k) + \Delta u(k)' S \Delta u(k)) + (1/2) x(N)' P_f x(N)$$

in which $\Delta u(k) = u(k) - u(k-1)$. Show that you can augment the state to include $u(k-1)$ via

$$\tilde{x}(k) = \begin{bmatrix} x(k) \\ u(k-1) \end{bmatrix}$$

and reduce this new problem to the standard LQR with the cross term. What are $\tilde{A}$, $\tilde{B}$, $\tilde{Q}$, $\tilde{R}$, and $\tilde{M}$ for the augmented problem (Rao and Rawlings, 1999)?

Exercise 1.26: Existence, uniqueness and stability with the cross term

Consider the linear quadratic problem with system

$$x^+ = Ax + Bu \quad (1.58)$$

and infinite horizon cost function

$$V(x(0), \mathbf{u}) = (1/2) \sum_{k=0}^{\infty} x(k)' Q x(k) + u(k)' R u(k)$$

The existence, uniqueness and stability conditions for this problem are: (A, B) stabilizable, $Q \geq 0$, (A, Q) detectable, and $R > 0$. Consider the modified objective function with the cross term

$$V = (1/2) \sum_{k=0}^{\infty} x(k)' Q x(k) + u(k)' R u(k) + 2x(k)' M u(k) \quad (1.59)$$

- (a) Consider reparameterizing the input as

$$v(k) = u(k) + T x(k) \quad (1.60)$$

Choose T such that the cost function in x and v does not have a cross term, and express the existence, uniqueness and stability conditions for the transformed system. Goodwin and Sin (1984, p.251) discuss this procedure in the state estimation problem with nonzero covariance between state and output measurement noises.

- (b) Translate and simplify these to obtain the existence, uniqueness and stability conditions for the original system with cross term.

Exercise 1.27: Forecasting and variance increase or decrease

Given positive definite initial state variance $P(0)$ and process disturbance variance Q , the variance after forecasting one sample time was shown to be

$$P^-(1) = AP(0)A' + Q$$

- (a) If A is stable, is it true that $AP(0)A' < P(0)$? If so, prove it. If not, provide a counterexample.
- (b) If A is unstable, is it true that $AP(0)A' > P(0)$? If so, prove it. If not, provide a counterexample.
- (c) If the magnitudes of *all* the eigenvalues of A are unstable, is it true that $AP(0)A' > P(0)$? If so, prove it. If not, provide a counterexample.

Exercise 1.28: Convergence of MHE with zero prior weighting

Show that the simplest form of MHE defined in (1.32) and (1.33) is also a convergent estimator for an observable system. What restrictions on the horizon length N do you require for this result to hold?

Hint: you can solve the MHE optimization problem by inspection when there is no prior weighting of the data.

Exercise 1.29: Symmetry in regulation and estimation

In this exercise we display the symmetry of the backward DP recursion for regulation, and the forward DP recursion for estimation. In the regulation problem we solve at stage k

$$\min_{x,u} \ell(z, u) + V_k^0(x) \quad \text{s.t. } x = Az + Bu$$

In backward DP, x is the state at the current stage and z is the state at the previous stage. The stage cost and cost to go are given by

$$\ell(z, u) = (1/2)(z'Qz + u'Ru) \quad V_k^0(x) = (1/2)x'\Pi(k)x$$

and the optimal cost is $V_{k-1}^0(z)$ since z is the state at the previous stage.

In estimation we solve at stage k

$$\min_{x,w} \ell(z, w) + V_k^0(x) \quad \text{s.t. } z = Ax + w$$

In forward DP, x is the state at the current stage, z is the state at the next stage. The stage cost and arrival cost are given by

$$\ell(z, w) = (1/2)(\|y(k+1) - Cz\|_{R^{-1}}^2 + w'Q^{-1}w) \quad V_k^0(x) = (1/2)\|x - \hat{x}(k)\|_{P(k)-1}^2$$

and we wish to find $V_{k+1}^0(z)$ in the estimation problem.

(a) In the estimation problem, take the z term outside the optimization and solve

$$\min_{x,w} \frac{1}{2} \left(w'Q^{-1}w + (x - \hat{x}(k))'P(k)^{-1}(x - \hat{x}(k)) \right) \quad \text{s.t. } z = Ax + w$$

using the inverse form in Exercise 1.18, and show that the optimal cost is given by

$$V^0(z) = (1/2)(z - A\hat{x}(k))'(P^-(k+1))^{-1}(z - A\hat{x}(k)) \\ P^-(k+1) = AP(k)A' + Q$$

Add the z term to this cost using the third part of Example 1.1 and show that

$$V_{k+1}^0(z) = (1/2)(z - \hat{x}(k+1))'P^{-1}(k+1)(z - \hat{x}(k+1)) \\ P(k+1) = P^-(k+1) - P^-(k+1)C'(CP^-(k+1)C' + R)^{-1}CP^-(k+1) \\ \hat{x}(k+1) = A\hat{x}(k) + L(k+1)(y(k+1) - CA\hat{x}(k)) \\ L(k+1) = P^-(k+1)C'(CP^-(k+1)C' + R)^{-1}$$

(b) In the regulator problem, take the z term outside the optimization and solve the remaining two-term problem using the regulator form of Exercise 1.18. Then

add the z term and show that

$$\begin{aligned} V_{k-1}^0(z) &= (1/2) z' \Pi(k-1) z \\ \Pi(k-1) &= Q + A' \Pi(k) A - A' \Pi(k) B (B' \Pi(k) B + R)^{-1} B' \Pi(k) A \\ u^0(z) &= K(k-1) z \\ x^0(z) &= (A + BK(k-1)) z \\ K(k-1) &= -(B' \Pi(k) B + R)^{-1} B' \Pi(k) A \end{aligned}$$

This symmetry can be developed further if we pose an output tracking problem rather than zero state regulation problem in the regulator.

Exercise 1.30: Symmetry in the Riccati iteration

Show that the covariance before measurement $P^-(k+1)$ in estimation satisfies an identical iteration to the cost to go $\Pi(k-1)$ in regulation under the change of variables $P^- \rightarrow \Pi, A \rightarrow A', C \rightarrow B'$.

Exercise 1.31: Detectability and observability canonical forms

Consider the partitioned system

$$\begin{aligned} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^+ &= \begin{bmatrix} A_{11} & 0 \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ y &= \begin{bmatrix} C_1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

with (A_{11}, C_1) observable. This form is known as observability canonical form.

- Show that the system is *not* observable by checking the rank of the observability matrix.
- Show that the modes x_1 can be uniquely determined from a sequence of measurements, but the modes x_2 *cannot* be uniquely determined from the measurements. The states x_2 are termed the unobservable modes.
- If A_{22} is stable the system is termed *detectable*. Although not all modes can be observed, the unobservable modes are stable and decay to steady state.

The following lemma gives an equivalent condition for detectability.

Lemma 1.13 (Hautus lemma for detectability). *A system is detectable if and only if*

$$\text{rank} \begin{bmatrix} \lambda I - A \\ C \end{bmatrix} = n \quad \text{for all } |\lambda| \geq 1$$

Prove this lemma using Lemma 1.4 as the condition for observability.

Exercise 1.32: Estimator stability and detectable systems

Show that the least squares estimator given in (1.27) is stable for (A, C) *detectable* with $Q > 0$.

Exercise 1.33: Estimator stability and semidefinite state noise penalty

We wish to show that the least squares estimator is stable for (A, C) detectable and $Q \geq 0$, (A, Q) stabilizable.

- (a) Because Q^{-1} is not defined in this problem, the objective function defined in (1.26) requires modification. Show that the objective function with semidefinite $Q \geq 0$ can be converted into the following form

$$V(x(0), w(T)) = \frac{1}{2} \left(|x(0) - \bar{x}(0)|_{(P^-(0))^{-1}}^2 + \sum_{k=0}^{T-1} |w(k)|_{\tilde{Q}}^2 + \sum_{k=0}^T |y(k) - Cx(k)|_{R^{-1}}^2 \right)$$

in which

$$x^+ = Ax + Gw \quad \tilde{Q} > 0$$

Find expressions for $\tilde{Q}$ and G in terms of the original semidefinite Q . How are the dimension of $\tilde{Q}$ and G related to the rank of Q ?

- (b) What is the probabilistic interpretation of the state estimation problem with semidefinite Q ?
- (c) Show that (A, Q) stabilizable implies (A, G) stabilizable in the converted form.
- (d) Show that this estimator is stable for (A, C) detectable and (A, G) stabilizable with $\tilde{Q}, R > 0$.
- (e) Discuss what happens to the estimator's stability if Q is not positive semidefinite or (A, Q) is not stabilizable.

Exercise 1.34: Calculating mean and variance from data

We are sampling a real-valued scalar random variable $x(k) \in \mathbb{R}$ at time k . Assume the random variable comes from a distribution with mean $\bar{x}$ and variance P , and the samples at different times are statistically independent.

A colleague has suggested the following formulas for estimating the mean and variance from N samples

$$\hat{x}_N = \frac{1}{N} \sum_{j=1}^N x(j) \quad \hat{P}_N = \frac{1}{N} \sum_{j=1}^N (x(j) - \hat{x}_N)^2$$

- (a) Prove that the estimate of the mean is unbiased for all N , i.e., show that for all N

$$\mathcal{E}(\hat{x}_N) = \bar{x}$$

- (b) Prove that the estimate of the variance is not unbiased for any N , i.e., show that for all N

$$\mathcal{E}(\hat{P}_N) \neq P$$

- (c) Using the result above, provide an alternative formula for the variance estimate that is unbiased for all N . How large does N have to be before these two estimates of P are within 1%?

Exercise 1.35: Expected sum of squares

Given that a random variable x has mean m and covariance P , show that the expected sum of squares is given by the formula (Selby, 1973, p.138)

$$\mathcal{E}(x'Qx) = m'Qm + \text{tr}(QP)$$

The trace of a square matrix A , written $\text{tr}(A)$, is defined to be the sum of the diagonal elements

$$\text{tr}(A) := \sum_i A_{ii}$$

Exercise 1.36: Normal distribution

Given a normal distribution with scalar parameters m and σ

$$p_\xi(x) = \sqrt{\frac{1}{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{x-m}{\sigma}\right)^2\right] \quad (1.61)$$

By direct calculation, show that

(a)

$$\begin{aligned} \mathcal{E}(\xi) &= m \\ \text{var}(\xi) &= \sigma^2 \end{aligned}$$

(b) Show that the mean and the maximum likelihood are equal for the normal distribution. Draw a sketch of this result. The maximum likelihood estimate, $\hat{x}$, is defined as

$$\hat{x} := \arg \max_x p_\xi(x)$$

in which $\arg$ returns the solution to the optimization problem.

Exercise 1.37: Conditional densities are positive definite

We show in Example A.44 that if ξ and η are jointly normally distributed as

$$\begin{aligned} \begin{bmatrix} \xi \\ \eta \end{bmatrix} &\sim N(m, P) \\ &\sim N\left(\begin{bmatrix} m_x \\ m_y \end{bmatrix}, \begin{bmatrix} P_x & P_{xy} \\ P_{yx} & P_y \end{bmatrix}\right) \end{aligned}$$

then the conditional density of ξ given η is also normal

$$(\xi|\eta) \sim N(m_{x|y}, P_{x|y})$$

in which the conditional mean is

$$m_{x|y} = m_x + P_{xy}P_y^{-1}(y - m_y)$$

and the conditional covariance is

$$P_{x|y} = P_x - P_{xy}P_y^{-1}P_{yx}$$

Given that the joint density is well defined, prove the marginal densities and the conditional densities also are well defined, i.e., given $P > 0$, prove $P_x > 0$, $P_y > 0$, $P_{x|y} > 0$, $P_{y|x} > 0$.

Exercise 1.38: Expectation and covariance under linear transformations

Consider the random variable $x \in \mathbb{R}^n$ with density p_x and mean and covariance

$$\mathcal{E}(x) = m_x \quad \text{cov}(x) = P_x$$

Consider the random variable $y \in \mathbb{R}^p$ defined by the linear transformation

$$y = Cx$$

- (a) Show that the mean and covariance for y are given by

$$\mathcal{E}(y) = Cm_x \quad \text{cov}(y) = CP_x C'$$

Does this result hold for all C ? If yes, prove it; if no, provide a counterexample.

- (b) Apply this result to solve Exercise A.35.

Exercise 1.39: Normal distributions under linear transformations

Given the normally distributed random variable, $\xi \in \mathbb{R}^n$, consider the random variable, $\eta \in \mathbb{R}^n$, obtained by the linear transformation

$$\eta = A\xi$$

in which A is a nonsingular matrix. Using the result on transforming probability densities, show that if $\xi \sim N(m, P)$, then $\eta \sim N(Am, APA')$. This result basically says that linear transformations of normal random variables are normal.

Exercise 1.40: More on normals and linear transformations

Consider a normally distributed random variable $x \in \mathbb{R}^n$, $x \sim N(m_x, P_x)$. You showed in Exercise 1.39 for $C \in \mathbb{R}^{n \times n}$ invertible, that the random variable y defined by the linear transformation $y = Cx$ is also normal and is distributed as

$$y \sim N(Cm_x, CP_x C')$$

Does this result hold for all C ? If yes, prove it; if no, provide a counterexample.

Exercise 1.41: Signal processing in the good old days—recursive least squares

Imagine we are sent back in time to 1960 and the only computers available have extremely small memories. Say we have a large amount of data coming from a process and we want to compute the least squares estimate of model parameters from these data. Our immediate challenge is that we cannot load all of these data into memory to make the standard least squares calculation.

Alternatively, go 150 years further back in time and consider the situation from Gauss's perspective,

It occasionally happens that after we have completed all parts of an extended calculation on a sequence of observations, we learn of a new observation that we would like to include. In many cases we will not want to have to redo the entire elimination but instead to find the modifications due to the new observation in the most reliable values of the unknowns and in their weights.

C.F. Gauss, 1823

G.W. Stewart Translation, 1995, p. 191.

Given the linear model

$$y_i = X'_i \theta$$

in which scalar y_i is the measurement at sample i , X'_i is the independent model variable (row vector, $1 \times p$) at sample i , and θ is the parameter vector ($p \times 1$) to be estimated from these data. Given the weighted least squares objective and n measurements, we wish to compute the usual estimate

$$\hat{\theta} = (X'X)^{-1}X'y \quad (1.62)$$

in which

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad X = \begin{bmatrix} X'_1 \\ \vdots \\ X'_n \end{bmatrix}$$

We do not wish to store the large matrices $X(n \times p)$ and $y(n \times 1)$ required for this calculation. Because we are planning to process the data one at a time, we first modify our usual least squares problem to deal with small n . For example, we wish to estimate the parameters when $n < p$ and the inverse in (1.62) does not exist. In such cases, we may choose to regularize the problem by modifying the objective function as follows

$$\Phi(\theta) = (\theta - \bar{\theta})' P_0^{-1} (\theta - \bar{\theta}) + \sum_{i=1}^n (y_i - X'_i \theta)^2$$

in which $\bar{\theta}$ and P_0 are chosen by the user. In Bayesian estimation, we call $\bar{\theta}$ and P_0 the prior information, and often assume that the prior density of θ (without measurements) is normal

$$\theta \sim N(\bar{\theta}, P_0)$$

The solution to this modified least squares estimation problem is

$$\hat{\theta} = \bar{\theta} + (X'X + P_0^{-1})^{-1}X'(y - X\bar{\theta}) \quad (1.63)$$

Devise a means to *recursively* estimate θ so that:

1. We never store more than one measurement at a time in memory.
2. After processing all the measurements, we obtain the same least squares estimate given in (1.63).

Exercise 1.42: Least squares parameter estimation and Bayesian estimation

Consider a model linear in the parameters

$$y = X\theta + e \quad (1.64)$$

in which $y \in \mathbb{R}^p$ is a vector of measurements, $\theta \in \mathbb{R}^m$ is a vector of parameters, $X \in \mathbb{R}^{p \times m}$ is a matrix of known constants, and $e \in \mathbb{R}^p$ is a random variable modeling the measurement error. The standard parameter estimation problem is to find the best estimate of θ given the measurements y corrupted with measurement error e , which we assume is distributed as

$$e \sim N(0, R)$$

- (a) Consider the case in which the errors in the measurements are independently and identically distributed with variance σ^2 , $R = \sigma^2 I$. For this case, the classic least squares problem and solution are

$$\min_{\theta} \|y - X\theta\|^2 \quad \hat{\theta} = (X'X)^{-1}X'y$$

Consider the measurements to be sampled from (1.64) with true parameter value θ_0 . Show that using the least squares formula, the parameter estimate is distributed as

$$\hat{\theta} \sim N(\theta_0, P_{\hat{\theta}}) \quad P_{\hat{\theta}} = \sigma^2 (X'X)^{-1}$$

- (b) Now consider again the model of (1.64) and a Bayesian estimation problem. Assume a prior distribution for the random variable θ

$$\theta \sim N(\bar{\theta}, \bar{P})$$

Compute the conditional density of θ given measurement y , show that this density is normal, and find its mean and covariance

$$p_{\theta|y}(\theta|y) = n(\theta, m, P)$$

Show that Bayesian estimation and least squares estimation give the same result in the limit of an infinite variance prior. In other words, if the covariance of the prior is large compared to the covariance of the measurement error, show that

$$m \approx (X'X)^{-1}X'y \quad P \approx P_{\hat{\theta}}$$

- (c) What (weighted) least squares minimization problem is solved for the general measurement error covariance

$$e \sim N(0, R)$$

Derive the least squares estimate formula for this case.

- (d) Again consider the measurements to be sampled from (1.64) with true parameter value θ_0 . Show that the weighted least squares formula gives parameter estimates that are distributed as

$$\hat{\theta} \sim N(\theta_0, P_{\hat{\theta}})$$

and find $P_{\hat{\theta}}$ for this case.

- (e) Show again that Bayesian estimation and least squares estimation give the same result in the limit of an infinite variance prior.

Exercise 1.43: Least squares and minimum variance estimation

Consider again the model linear in the parameters and the least squares estimator from Exercise 1.42

$$y = X\theta + e \quad e \sim N(0, R)$$

$$\hat{\theta} = (X'R^{-1}X)^{-1}X'R^{-1}y$$

Show that the covariance of the least squares estimator is the smallest covariance of all linear unbiased estimators.

Exercise 1.44: Two stages are not better than one

We often can decompose an estimation problem into stages. Consider the following case in which we wish to estimate x from measurements of z , but we have the model between x and an intermediate variable, y , and the model between y and z

$$y = Ax + e_1 \quad \text{cov}(e_1) = Q_1$$

$$z = By + e_2 \quad \text{cov}(e_2) = Q_2$$

- (a) Write down the optimal least squares problem to solve for $\hat{y}$ given the z measurements and the second model. Given $\hat{y}$, write down the optimal least squares problem for $\hat{x}$ in terms of $\hat{y}$. Combine these two results together and write the resulting estimate of $\hat{x}$ given measurements of z . Call this the two-stage estimate of x .

- (b) Combine the two models together into a single model and show that the relationship between z and x is

$$z = BAx + e_3 \quad \text{cov}(e_3) = Q_3$$

Express Q_3 in terms of Q_1, Q_2 and the models A, B . What is the optimal least squares estimate of $\hat{x}$ given measurements of z and the one-stage model? Call this the one-stage estimate of x .

- (c) Are the one-stage and two-stage estimates of x the same? If yes, prove it. If no, provide a counterexample. Do you have to make any assumptions about the models A, B ?

Exercise 1.45: Time-varying Kalman filter

Derive formulas for the conditional densities of $x(k)|y(k-1)$ and $x(k)|y(k)$ for the time-varying linear system

$$\begin{aligned} x(k+1) &= A(k)x(k) + G(k)w(k) \\ y(k) &= C(k)x(k) + v(k) \end{aligned}$$

in which the initial state, state noise and measurement noise are independently distributed as

$$x(0) \sim N(\bar{x}_0, Q_0) \quad w(k) \sim N(0, Q) \quad v(k) \sim N(0, R)$$

Exercise 1.46: More on conditional densities

In deriving the discrete time Kalman filter, we have $p_{x|y}(x(k)|y(k))$ and we wish to calculate recursively $p_{x|y}(x(k+1)|y(k+1))$ after we collect the output measurement at time $k+1$. It is straightforward to calculate $p_{x,y|y}(x(k+1), y(k+1)|y(k))$ from our established results on normal densities and knowledge of $p_{x|y}(x(k)|y(k))$, but we still need to establish a formula for pushing the $y(k+1)$ to the other side of the conditional density bar. Consider the following statement as a possible lemma to aid in this operation.

$$p_{a|b,c}(a|b, c) = \frac{p_{a,b|c}(a, b|c)}{p_{b|c}(b|c)}$$

If this statement is true, prove it. If it is false, give a counterexample.

Exercise 1.47: Other useful conditional densities

Using the definitions of marginal and conditional density, establish the following useful conditional density relations

1. $p_{A|B}(a|b) = \int p_{A|B,C}(a|b, c) p_{C|B}(c|b) dc$
2. $p_{A|B,C}(a|b, c) = p_{C|A,B}(c|a, b) \frac{p_{A|B}(a|b)}{p_{C|B}(c|b)}$

Exercise 1.48: Optimal filtering and deterministic least squares

Given the data sequence $(y(0), \dots, y(k))$ and the system model

$$x^+ = Ax + w$$

$$y = Cx + v$$

- (a) Write down a least squares problem whose solution would provide a good state estimate for $x(k)$ in this situation. What probabilistic interpretation can you assign to the estimate calculated from this least squares problem?
- (b) Now consider the nonlinear model

$$x^+ = f(x) + w$$

$$y = g(x) + v$$

What is the corresponding nonlinear least squares problem for estimating $x(k)$ in this situation? What probabilistic interpretation, if any, can you assign to this estimate in the nonlinear model context?

- (c) What is the motivation for changing from these least squares estimators to the moving horizon estimators we discussed in the chapter?

Exercise 1.49: A nonlinear transformation and conditional density

Consider the following relationship between the random variable y , and x and v

$$y = f(x) + v$$

The author of a famous textbook wants us to believe that

$$p_{y|x}(y|x) = p_v(y - f(x))$$

Derive this result and state what additional assumptions on the random variables x and v are required for this result to be correct.

Exercise 1.50: Some smoothing

One of the problems with asking you to derive the Kalman filter is that the derivation is in so many textbooks that it is difficult to tell if you are thinking independently. So here's a variation on the theme that should help you evaluate your level of understanding of these ideas. Let's calculate a smoothed rather than filtered estimate and covariance. Here's the problem.

We have the usual setup with a prior on $x(0)$

$$x(0) \sim N(\bar{x}(0), Q_0)$$

and we receive data from the following system

$$x(k+1) = Ax(k) + w(k)$$

$$y(k) = Cx(k) + v(k)$$

in which the random variables $w(k)$ and $v(k)$ are independent, identically distributed normals, $w(k) \sim N(0, Q)$, $v(k) \sim N(0, R)$.

- (a) Calculate the standard density for the filtering problem, $p_{x(0)|y(0)}(x(0)|y(0))$.

- (b) Now calculate the density for the smoothing problem

$$p_{x(0)|y(0),y(1)}(x(0)|y(0),y(1))$$

that is, *not* the usual $p_{x(1)|y(0),y(1)}(x(1)|y(0),y(1))$.

Exercise 1.51: Alive on arrival

The following two optimization problems are helpful in understanding the arrival cost decomposition in state estimation.

- (a) Let $V(x, y, z)$ be a positive, strictly convex function consisting of the sum of two functions, one of which depends on both x and y , and the other of which depends on y and z

$$V(x, y, z) = g(x, y) + h(y, z) \quad V : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R}^p \rightarrow \mathbb{R}_{\geq 0}$$

Consider the optimization problem

$$P1 : \min_{x,y,z} V(x, y, z)$$

The arrival cost decomposes this three-variable optimization problem into two, smaller dimensional optimization problems. Define the “arrival cost” $\tilde{g}$ for this problem as the solution to the following single-variable optimization problem

$$\tilde{g}(y) = \min_x g(x, y)$$

and define optimization problem $P2$ as follows

$$P2 : \min_{y,z} \tilde{g}(y) + h(y, z)$$

Let (x', y', z') denote the solution to $P1$ and (x^0, y^0, z^0) denote the solution to $P2$, in which

$$x^0 = \arg \min_x g(x, y^0)$$

Prove that the two solutions are equal

$$(x', y', z') = (x^0, y^0, z^0)$$

- (b) Repeat the previous part for the following optimization problems

$$V(x, y, z) = g(x) + h(y, z)$$

Here the y variables do not appear in g but restrict the x variables through a linear constraint. The two optimization problems are

$$P1 : \min_{x,y,z} V(x, y, z) \quad \text{subject to } Ex = y$$

$$P2 : \min_{y,z} \tilde{g}(y) + h(y, z)$$

in which

$$\tilde{g}(y) = \min_x g(x) \quad \text{subject to } Ex = y$$

Exercise 1.52: On-time arrival

Consider the deterministic, full information state estimation optimization problem

$$\min_{x(0), \mathbf{w}, \mathbf{v}} \frac{1}{2} \left(|x(0) - \bar{x}(0)|_{(P^-(0))^{-1}}^2 + \sum_{i=0}^{T-1} |w(i)|_{Q^{-1}}^2 + |v(i)|_{R^{-1}}^2 \right) \quad (1.65)$$

subject to

$$\begin{aligned} x^+ &= Ax + w \\ y &= Cx + v \end{aligned} \quad (1.66)$$

in which the sequence of measurements $\mathbf{y}(T)$ are known values. Notice we assume the noise-shaping matrix, G , is an identity matrix here. See Exercise 1.53 for the general case. Using the result of the first part of Exercise 1.51, show that this problem is equivalent to the following problem

$$\min_{x(T-N), \mathbf{w}, \mathbf{v}} V_{T-N}^-(x(T-N)) + \frac{1}{2} \sum_{i=T-N}^{T-1} |w(i)|_{Q^{-1}}^2 + |v(i)|_{R^{-1}}^2$$

subject to (1.66). The arrival cost is defined as

$$V_N^-(a) := \min_{x(0), \mathbf{w}, \mathbf{v}} \frac{1}{2} \left(|x(0) - \bar{x}(0)|_{(P^-(0))^{-1}}^2 + \sum_{i=0}^{N-1} |w(i)|_{Q^{-1}}^2 + |v(i)|_{R^{-1}}^2 \right)$$

subject to (1.66) and $x(N) = a$. Notice that any value of N , $0 \leq N \leq T$, can be used to split the cost function using the arrival cost.

Exercise 1.53: Arrival cost with noise-shaping matrix G

Consider the deterministic, full information state estimation optimization problem

$$\min_{x(0), \mathbf{w}, \mathbf{v}} \frac{1}{2} \left(|x(0) - \bar{x}(0)|_{(P^-(0))^{-1}}^2 + \sum_{i=0}^{T-1} |w(i)|_{Q^{-1}}^2 + |v(i)|_{R^{-1}}^2 \right)$$

subject to

$$\begin{aligned} x^+ &= Ax + Gw \\ y &= Cx + v \end{aligned} \quad (1.67)$$

in which the sequence of measurements $\mathbf{y}$ are known values. Using the result of the second part of Exercise 1.51, show that this problem also is equivalent to the following problem

$$\min_{x(T-N), \mathbf{w}, \mathbf{v}} V_{T-N}^-(x(T-N)) + \frac{1}{2} \left(\sum_{i=T-N}^{T-1} |w(i)|_{Q^{-1}}^2 + |v(i)|_{R^{-1}}^2 \right)$$

subject to (1.67). The arrival cost is defined for all $k \geq 0$ and $a \in \mathbb{R}^n$ by

$$V_k^-(a) := \min_{x(0), \mathbf{w}, \mathbf{v}} \frac{1}{2} \left(|x(0) - \bar{x}(0)|_{(P^-(0))^{-1}}^2 + \sum_{i=0}^{k-1} |w(i)|_{Q^{-1}}^2 + |v(i)|_{R^{-1}}^2 \right)$$

subject to $x(k) = a$ and the model (1.67). Notice that any value of N , $0 \leq N \leq T$, can be used to split the cost function using the arrival cost.

Exercise 1.54: Where is the steady state?

Consider the two-input, two-output system

$$A = \begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0.6 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.6 \end{bmatrix} \quad B = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.4 \\ 0.25 & 0 \\ 0 & 0.6 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

- The output setpoint is $y_{sp} = [1 \quad -1]'$ and the input setpoint is $u_{sp} = [0 \quad 0]'$. Calculate the target triple (x_s, u_s, y_s) . Is the output setpoint feasible, i.e., does $y_s = y_{sp}$?
- Assume only input one u_1 is available for control. Is the output setpoint feasible? What is the target in this case using $Q_s = I$?
- Assume both inputs are available for control but only the first output has a setpoint, $y_{1t} = 1$. What is the solution to the target problem for $R_s = I$?

Exercise 1.55: Detectability of integrating disturbance models

- Prove Lemma 1.8; the augmented system is detectable if and only if the system (A, C) is detectable and

$$\text{rank} \begin{bmatrix} I - A & -B_d \\ C & C_d \end{bmatrix} = n + n_d$$

- Prove Corollary 1.9; the augmented system is detectable only if $n_d \leq p$.

Exercise 1.56: Unconstrained tracking problem

- For an *unconstrained* system, show that the following condition is *sufficient* for feasibility of the target problem for any r_{sp} .

$$\text{rank} \begin{bmatrix} I - A & -B \\ HC & 0 \end{bmatrix} = n + n_c \quad (1.68)$$

- Show that (1.68) implies that the number of controlled variables without offset is less than or equal to the number of manipulated variables and the number of measurements, $n_c \leq m$ and $n_c \leq p$.
- Show that (1.68) implies the rows of H are independent.
- Does (1.68) imply that the rows of C are independent? If so, prove it; if not, provide a counterexample.
- By choosing H , how can one satisfy (1.68) if one has installed redundant sensors so several rows of C are identical?

Exercise 1.57: Unconstrained tracking problem for stabilizable systems

If we restrict attention to stabilizable systems, the sufficient condition of Exercise 1.56 becomes a necessary and sufficient condition. Prove the following lemma.

Lemma 1.14 (Stabilizable systems and feasible targets). *Consider an unconstrained, stabilizable system (A, B) . The target is feasible for any r_{sp} if and only if*

$$\text{rank} \begin{bmatrix} I - A & -B \\ HC & 0 \end{bmatrix} = n + n_c$$

Exercise 1.58: Existence and uniqueness of the unconstrained target

Assume a system having p controlled variables $z = Hx$, with setpoints r_{sp} , and m manipulated variables u , with setpoints u_{sp} . Consider the steady-state target problem

$$\min_{x, u} (1/2)(u - u_{sp})' R (u - u_{sp}) \quad R > 0$$

subject to

$$\begin{bmatrix} I - A & -B \\ H & 0 \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} = \begin{bmatrix} 0 \\ r_{sp} \end{bmatrix}$$

Show that the steady-state solution (x, u) exists for any (r_{sp}, u_{sp}) and is unique if

$$\text{rank} \begin{bmatrix} I - A & -B \\ H & 0 \end{bmatrix} = n + p \quad \text{rank} \begin{bmatrix} I - A \\ H \end{bmatrix} = n$$

Exercise 1.59: Choose a sample time

Consider the unstable continuous time system

$$\frac{dx}{dt} = Ax + Bu \quad y = Cx$$

in which

$$A = \begin{bmatrix} -0.281 & 0.935 & 0.035 & 0.008 \\ 0.047 & -0.116 & 0.053 & 0.383 \\ 0.679 & 0.519 & 0.030 & 0.067 \\ 0.679 & 0.831 & 0.671 & -0.083 \end{bmatrix} \quad B = \begin{bmatrix} 0.687 \\ 0.589 \\ 0.930 \\ 0.846 \end{bmatrix} \quad C = I$$

Consider regulator tuning parameters and constraints

$$Q = \text{diag}(1, 2, 1, 2) \quad R = 1 \quad N = 10 \quad |x| \leq \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$$

- (a) Compute the eigenvalues of A . Choose a sample time of $\Delta = 0.04$ and simulate the MPC regulator response given $x(0) = [-0.9 \quad -1.8 \quad 0.7 \quad 2]'$ until $t = 20$. Use an ODE solver to simulate the continuous time plant response. Plot all states and the input versus time.

Now add an input disturbance to the regulator so the control applied to the plant is u_d instead of u in which

$$u_d(k) = (1 + 0.1w_1)u(k) + 0.1w_2$$

and w_1 and w_2 are zero-mean, normally distributed random variables with unit variance. Simulate the regulator's performance given this disturbance. Plot all states and $u_d(k)$ versus time.

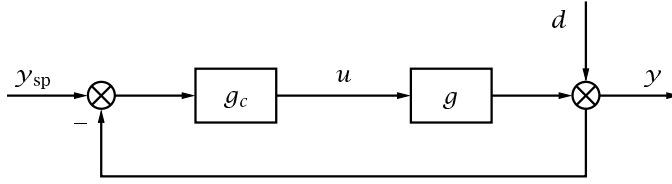


Figure 1.14: Feedback control system with output disturbance d , and setpoint y_{sp} .

- (b) Repeat the simulations with and without disturbance for $\Delta = 0.4$ and $\Delta = 2$.
- (c) Compare the simulations for the different sample times. What happens if the sample time is too large? Choose an appropriate sample time for this system and justify your choice.

Exercise 1.60: Disturbance models and offset

Consider the following two-input, three-output plant discussed in Example 1.11

$$\begin{aligned} x^+ &= Ax + Bu + B_p p \\ y &= Cx \end{aligned}$$

in which

$$A = \begin{bmatrix} 0.2681 & -0.00338 & -0.00728 \\ 9.703 & 0.3279 & -25.44 \\ 0 & 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} -0.00537 & 0.1655 \\ 1.297 & 97.91 \\ 0 & -6.637 \end{bmatrix} \quad B_p = \begin{bmatrix} -0.1175 \\ 69.74 \\ 6.637 \end{bmatrix}$$

The input disturbance p results from a reactor inlet flowrate disturbance.

- (a) Since there are two inputs, choose two outputs in which to remove steady-state offset. Build an output disturbance model with two integrators. Is your augmented model detectable?
- (b) Implement your controller using $p = 0.01$ as a step disturbance at $k = 0$. Do you remove offset in your chosen outputs? Do you remove offset in any outputs?
- (c) Can you find any two-integrator disturbance model that removes offset in two outputs? If so, which disturbance model do you use? If not, why not?

Exercise 1.61: MPC, PID, and time delay

Consider the following first-order system with time delay shown in Figure 1.14

$$g(s) = \frac{k}{\tau s + 1} e^{-\theta s}, \quad k = 1, \tau = 1, \theta = 5$$

Consider a unit step change in setpoint y_{sp} , at $t = 0$.

- (a) Choose a reasonable sample time, Δ , and disturbance model, and simulate an offset-free discrete time MPC controller for this setpoint change. List all of your chosen parameters.
- (b) Choose PID tuning parameters to achieve “good performance” for this system. List your PID tuning parameters. Compare the performances of the two controllers.

Exercise 1.62: CSTR heat-transfer coefficient

Your mission is to design the controller for the nonlinear CSTR model given in Example 1.11. We wish to use a linear controller and estimator with three integrating disturbances to remove offset in two controlled variables: *temperature* and *level*; use the nonlinear CSTR model as the plant.

- (a) You are particularly concerned about disturbances to the heat-transfer rate (parameter U) for this reactor. If changes to U are the primary disturbance, what disturbance model do you recommend and what covariances do you recommend for the three disturbances so that the disturbance state accounting for heat transfer is used primarily to explain the output error in the state estimator? First do a simulation with no measurement noise to test your estimator design. In the simulation let the reactor's heat-transfer coefficient decrease (and increase) by 20% at 10 minutes to test your control system design. Comment on the performance of the control system.
- (b) Now let's add some measurement noise to all three sensors. So we all work on the same problem, choose the variance of the measurement error R_v to be

$$R_v = 10^{-3} \text{diag}(c_s^2, T_s^2, h_s^2)$$

in which (c_s, T_s, h_s) are the nominal steady states of the three measurements. Is the performance from the previous part assuming no measurement noise acceptable? How do you adjust your estimator from the previous part to obtain good performance? Rerun the simulation with measurement noise and your adjusted state estimator. Comment on the change in the performance of your new design that accounts for the measurement noise.

- (c) Recall that the offset lemma 1.10 is an either-or proposition, i.e., *either* the controller removes steady offset in the controlled variables *or* the system is closed-loop unstable. From closed-loop simulation, approximate the range of plant U values for which the controller is stabilizing (with zero measurement noise). From a stabilization perspective, which disturbance is worse, an increase or decrease in the plant's heat-transfer coefficient?

Exercise 1.63: System identification of the nonlinear CSTR

In many practical applications, it may not be convenient to express system dynamics from first principles. Hence, identifying a suitable model from data is a critical step in the design of an MPC controller. Your final mission is to obtain a 2-input, 3-output process model for the nonlinear CSTR given in Example 1.11 using the System Identification Toolbox in MATLAB. Relevant functions are provided.

- (a) Begin first by creating a dataset for identification. Generate a pseudo-random, binary signal (PRBS) for the inputs using `idinput`. Ensure you have generated

uncorrelated signals for each input. Think about the amplitude of the PRBS to use when collecting data from a nonlinear process keeping in mind that large perturbations may lead to undesirable phenomena such as reactor ignition. Inject these generated input sequences into the nonlinear plant of Example 1.11 and simulate the system by solving the nonlinear ODEs. Add measurement noise to the simulation so that you have a realistic dataset for the ID and plot the input-output data.

- (b) Use the data to identify a third-order linear state-space model by calling `iddata` and `ssest`. Compare the step tests of your identified model with those from the linear model used in Example 1.11. Which is more accurate compared to the true plant simulation?
- (c) Using the code for Example 1.11 as a starting point, replace the linear model in the MPC controller with your identified model and recalculate Figures 1.10 and 1.11 from the example. Is your control system robust enough to obtain good closed-loop control of the nonlinear plant using your linear model identified from data in the MPC controller? Do you maintain zero offset in the controlled variables?

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